

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

19ECE73 – CRYPTOGRAPHY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

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|-------|---|----|-----|---|---|
| 1. a) | Consider a 26 letter alphabet system. Use One Time Pad Vigenère cipher. | | | | |
| i. | Encrypt the plaintext "sendmoremoney" with the key stream 9 0 1 7 23 15 21 14 11 11 2 8 9. | | | | |
| ii. | Using the ciphertext produced in part a, find a key so that the cipher text decrypts to the plaintext "cashnotneeded" | 10 | L*3 | 1 | 1 |
| b) | Using Euclidean algorithm, represent $d = u \times 826 + v \times 2789$ where $d = \text{g.c.d}(826, 2789)$ | 10 | L3 | 2 | 1 |
| 2. a) | Construct a playfair matrix using the key: DECRYPTION. Using the same decrypt the following cipher text:
ONTSDYRBYCOVNBBPNFBUTLVPNPTVFM TBGQ TWCBGOSC
RIZPOERZ | 10 | L3 | 1 | 1 |
| b) | Find the smallest non-negative solution to the following congruences.
$y \equiv 2 \pmod{3}$
$y \equiv 3 \pmod{5}$
$y \equiv 4 \pmod{11}$
$y \equiv 5 \pmod{16}$ | 10 | L3 | 2 | 1 |
| 3. a) | Suppose that we know that our adversary is using a 2×2 enciphering matrix with a 29-letter alphabet, where A-Z have the usual numerical equivalents, blank()=26, ?=27, !=28. We receive the message "GFPYJP_X?UYXSTLADPLW", and we suppose that we know that the last five letters of plaintext are our adversary's signature "KARLA". Decrypt the information using Hill Cipher. | 10 | L4 | 1 | 2 |
| b) | Decrypt the following cipher and answer the question.
HIJMRIDHMMWGVIXIDPSKEVMXLQ
Consider substitution cipher with key =4. There are 27 letters in the alphabet. A - Z as 0 - 25 and _ as 26. | 10 | L4 | 2 | 2 |

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | With the help of a neat flow diagram, explain Feistel Cipher Structure. What are the parameters considered in the design of a Fiestal Cipher? | 10 | L2 | 3 | 1 |
| b) | Generate the cipher text for the information "ME" using RSA algorithm. Consider $p=5$ and $q=11$. Also show the decryption process of the encrypted message. Choose $e = 7$. | 10 | L3 | 4 | 1 |

P.T.O.

5. a) Consider a Diffie-Hellman scheme with a common prime $q = 11$ and a primitive root $\alpha = 2$.
- Show that 2 is a primitive root of 11.
 - If user A has public key $Y_A = 9$, what is A's private key X_A ?
 - If user B has public key $Y_B = 3$, what is its private key?
 - What is the shared secret key K ?
- b) List and explain seven criteria for S box design in Data Encryption Standard.
6. a) Explain Simple Secret Key Distribution Algorithm. How this scheme can be modified to ensure confidentiality and authentication?
- b) Plot the Elliptic Curve $E(1,1)$ and hence show the addition of two points on the curve.

10	L3	3	1
10	L2	4	1
10	L2	3	1
10	L3	4	1

Unit – III

7. a) With the help of neat block diagrams explain the variants of Hash Functions.
- b) What are digital signatures? Explain and compare direct and arbitrated digital signature.
8. a) Compare and contrast the principles used in MAC and Hash functions in assuring authentication.
- b) What is Quantum Cryptography? Explain.

10	L2	5	1
10	L2	5	1
10	L2	5	1
10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
